

Run report

2026-09-11T023002Z

Configuration

Parameter	Value	Parameter	Value
batch_size	1,024	n_samples	1,600,000
dim	12	dataset	jets
seed	799,905,857	data_seed	42
hidden_units	128	n_layers	3
n_epochs	100	n_disc_steps	5
lr_g	3.000×10^{-5}	lr_d	1.000×10^{-4}
lambda_dispersion	0.01500	mmd_subsample	16,384
mmd_test	2.000×10^{-4}	best_epoch	23
compile_cache_warm	True		
variables	m [GeV], $\ln \rho$, $\lambda_{0.5}^1$, w , λ_2^1 , z_g , $\tau_{21}^{(\beta=1)}$, M , n_{ch} , f_{ch} , p_T^D , q		
mmd_sigmas_detector	$3.107 \times (\frac{1}{2}, \frac{1}{\sqrt{2}}, 1, \sqrt{2}, 2)$		
mmd_sigmas_particle	$3.027 \times (\frac{1}{2}, \frac{1}{\sqrt{2}}, 1, \sqrt{2}, 2)$		

Timing

Phase	Seconds	Share (%)	Detail
train	41.873	73.9	train
transfer	6.988		train
compile	3.748		train
epochs	29.801		train
select	1.132		train
particle_mmd	0.400	0.7	train
save	0.276	0.5	train
data	1.523	2.7	cache hit – load
load	2.957	5.2	load
plots	9.645	17.0	load
evaluate	0.005	0.0	load
total	56.679	100.0	

Detector-level metrics

Distance to Data. Improvements are relative to Simulation.

Wasserstein [$\times 10^3$]

Observable	Sim.	IBU	IBU impr. (%)	RAN	RAN impr. (%)
Mass and hard scale (IRC-safe kinematics)					
m [GeV]	208.9	138.8	33.6	5.682	97.3
$\ln \rho$	227.9	20.71	90.9	9.906	95.7
Continuous angularities (IRC-safe jet shapes)					
$\lambda_{0.5}^1$	262.6	12.84	95.1	6.161	97.7
w	265.6	9.978	96.2	5.614	97.9
λ_2^1	255.4	9.215	96.4	7.588	97.0
Splitting and 2-prong substructure					
$z_g^\dagger$	60.45	60.45	0.0	18.59	69.3
$\tau_{21}^{(\beta=1)}$	121.8	70.10	42.4	21.19	82.6
Hadronization, multiplicity and fragmentation (IRC-unsafe)					
M	224.2	177.3	20.9	9.860	95.6
n_{ch}	220.8	172.3	22.0	8.347	96.2
f_{ch}	52.86	40.94	22.6	5.788	89.1
p_T^D	256.8	30.80	88.0	10.13	96.1
q	79.67	17.51	78.0	3.693	95.4

[†] IBU's purity binning produced a single bin, so IBU failed to unfold.

JS divergence [$\times 10^3$]

Observable	Sim.	IBU	IBU impr. (%)	RAN	RAN impr. (%)
Mass and hard scale (IRC-safe kinematics)					
m [GeV]	6.762	3.666	45.8	0.07514	98.9
$\ln \rho$	8.383	0.4017	95.2	0.1476	98.2
Continuous angularities (IRC-safe jet shapes)					
$\lambda_{0.5}^1$	9.359	0.4714	95.0	0.1255	98.7
w	9.139	0.4527	95.0	0.1335	98.5
λ_2^1	7.776	0.1728	97.8	0.1344	98.3
Splitting and 2-prong substructure					
$z_g^\dagger$	0.6723	0.6723	0.0	0.1249	81.4
$\tau_{21}^{(\beta=1)}$	3.372	1.934	42.7	0.1932	94.3
Hadronization, multiplicity and fragmentation (IRC-unsafe)					
M	9.640	7.633	20.8	0.1237	98.7
n_{ch}	7.512	6.314	16.0	0.09658	98.7
f_{ch}	1.033	0.7960	22.9	0.09605	90.7
p_T^D	8.223	1.311	84.1	0.1103	98.7
q	1.576	0.2512	84.1	0.07596	95.2

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VLC divergence [$\times 10^3$]

Observable	Sim.	IBU	IBU impr. (%)	RAN	RAN impr. (%)
Mass and hard scale (IRC-safe kinematics)					
m [GeV]	26.93	14.62	45.7	0.2801	99.0
$\ln \rho$	33.31	1.576	95.3	0.5712	98.3
Continuous angularities (IRC-safe jet shapes)					
$\lambda_{0.5}^1$	37.19	1.877	95.0	0.4957	98.7
w	36.32	1.800	95.0	0.5257	98.6
λ_2^1	30.90	0.6583	97.9	0.4992	98.4
Splitting and 2-prong substructure					
$z_g^\dagger$	2.683	2.683	0.0	0.4997	81.4
$\tau_{21}^{(\beta=1)}$	13.44	7.712	42.6	0.7684	94.3
Hadronization, multiplicity and fragmentation (IRC-unsafe)					
M	38.25	30.31	20.8	0.4638	98.8
n_{ch}	29.81	25.07	15.9	0.3599	98.8
f_{ch}	4.125	3.177	23.0	0.3842	90.7
p_T^D	32.65	5.211	84.0	0.4367	98.7
q	6.261	0.9940	84.1	0.2948	95.3

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Particle-level metrics

Distance to Truth. Improvements are relative to Generation.

Wasserstein [$\times 10^3$]

Observable	Gen	IBU	IBU impr. (%)	RAN	RAN impr. (%)
Mass and hard scale (IRC-safe kinematics)					
m [GeV]	111.7	94.07	15.8	100.5	10.1
$\ln \rho$	253.1	32.57	87.1	20.97	91.7
Continuous angularities (IRC-safe jet shapes)					
$\lambda_{0.5}^1$	303.0	18.71	93.8	6.166	98.0
w	284.1	12.29	95.7	5.787	98.0
λ_2^1	253.8	11.13	95.6	4.581	98.2
Splitting and 2-prong substructure					
$z_g^\dagger$	113.4	113.4	0.0	25.92	77.1
$\tau_{21}^{(\beta=1)}$	259.2	150.5	41.9	13.69	94.7
Hadronization, multiplicity and fragmentation (IRC-unsafe)					
M	256.8	204.8	20.2	38.25	85.1
n_{ch}	234.6	188.6	19.6	42.50	81.9
f_{ch}	45.78	45.33	1.0	9.538	79.2
p_T^D	236.3	55.36	76.6	17.68	92.5
q	79.37	18.15	77.1	4.036	94.9

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JS divergence [$\times 10^3$]

Observable	Gen	IBU	IBU impr. (%)	RAN	RAN impr. (%)
Mass and hard scale (IRC-safe kinematics)					
m [GeV]	1.963	11.86	−504.3	1.884	4.0
$\ln \rho$	9.878	1.731	82.5	0.2049	97.9
Continuous angularities (IRC-safe jet shapes)					
$\lambda_{0.5}^1$	12.34	2.009	83.7	0.2146	98.3
w	11.39	1.465	87.1	0.2309	98.0
λ_2^1	8.824	0.8335	90.6	0.1672	98.1
Splitting and 2-prong substructure					
$z_g^\dagger$	2.041	2.041	0.0	0.1661	91.9
$\tau_{21}^{(\beta=1)}$	10.50	6.577	37.4	0.1879	98.2
Hadronization, multiplicity and fragmentation (IRC-unsafe)					
M	7.842	12.83	−63.7	0.3706	95.3
n_{ch}	6.447	9.773	−51.6	0.3425	94.7
f_{ch}	1.025	0.9435	7.9	0.1389	86.4
p_T^D	8.342	4.029	51.7	0.2449	97.1
q	2.027	0.9958	50.9	0.09582	95.3

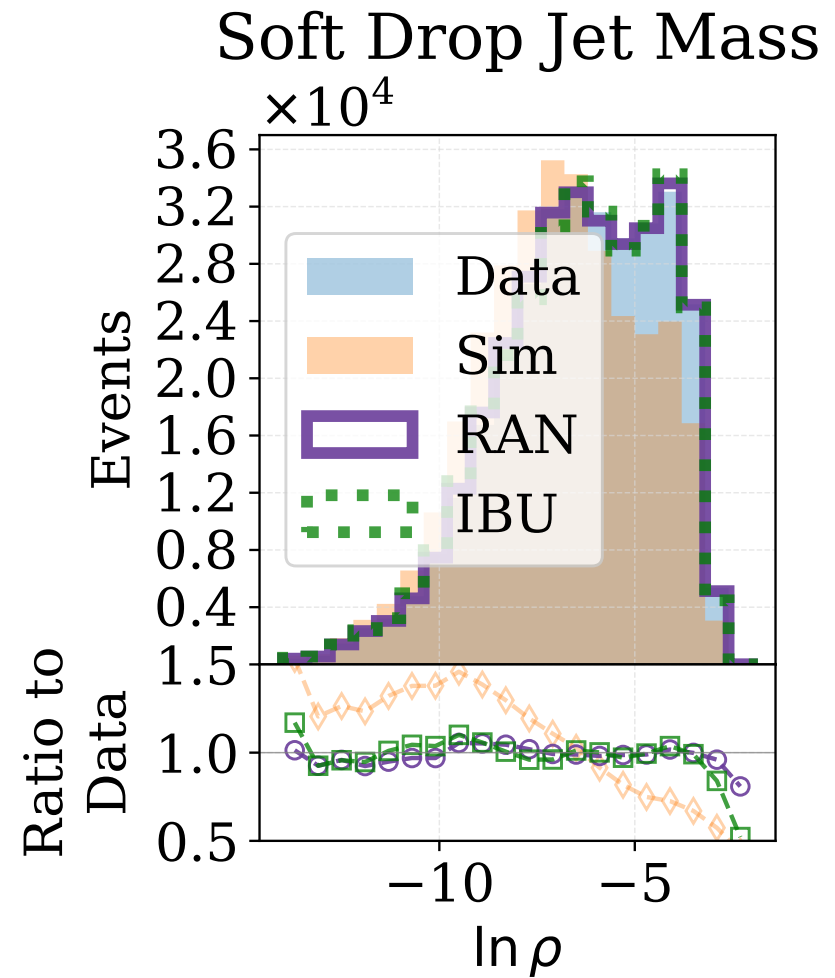
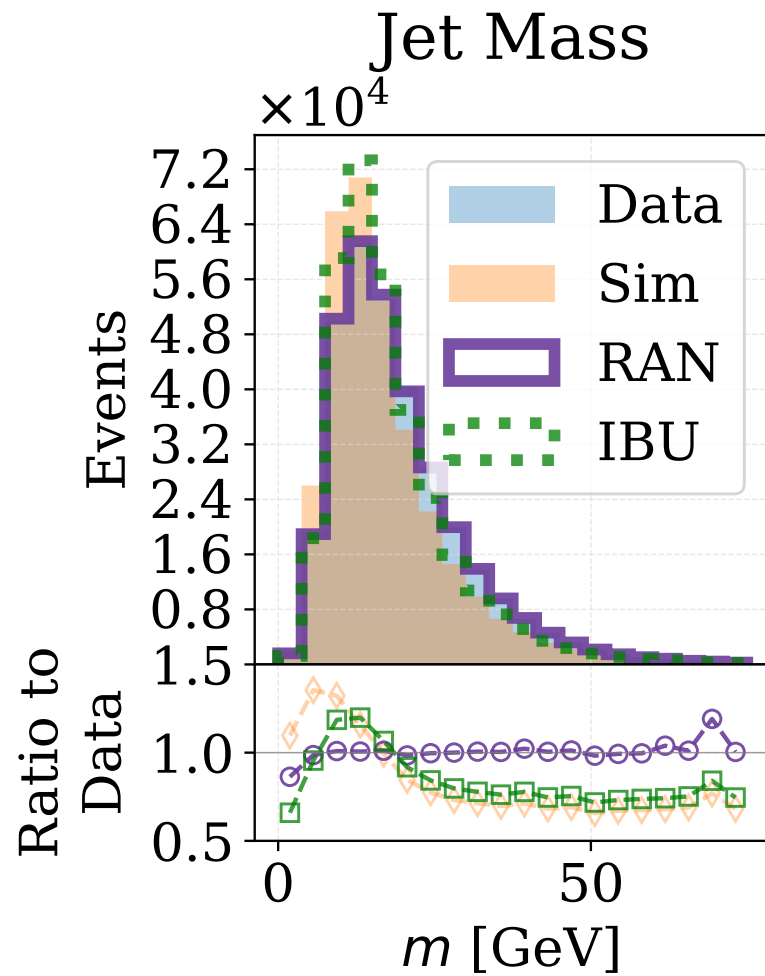
[†] IBU's purity binning produced a single bin, so IBU failed to unfold.

VLC divergence [$\times 10^3$]

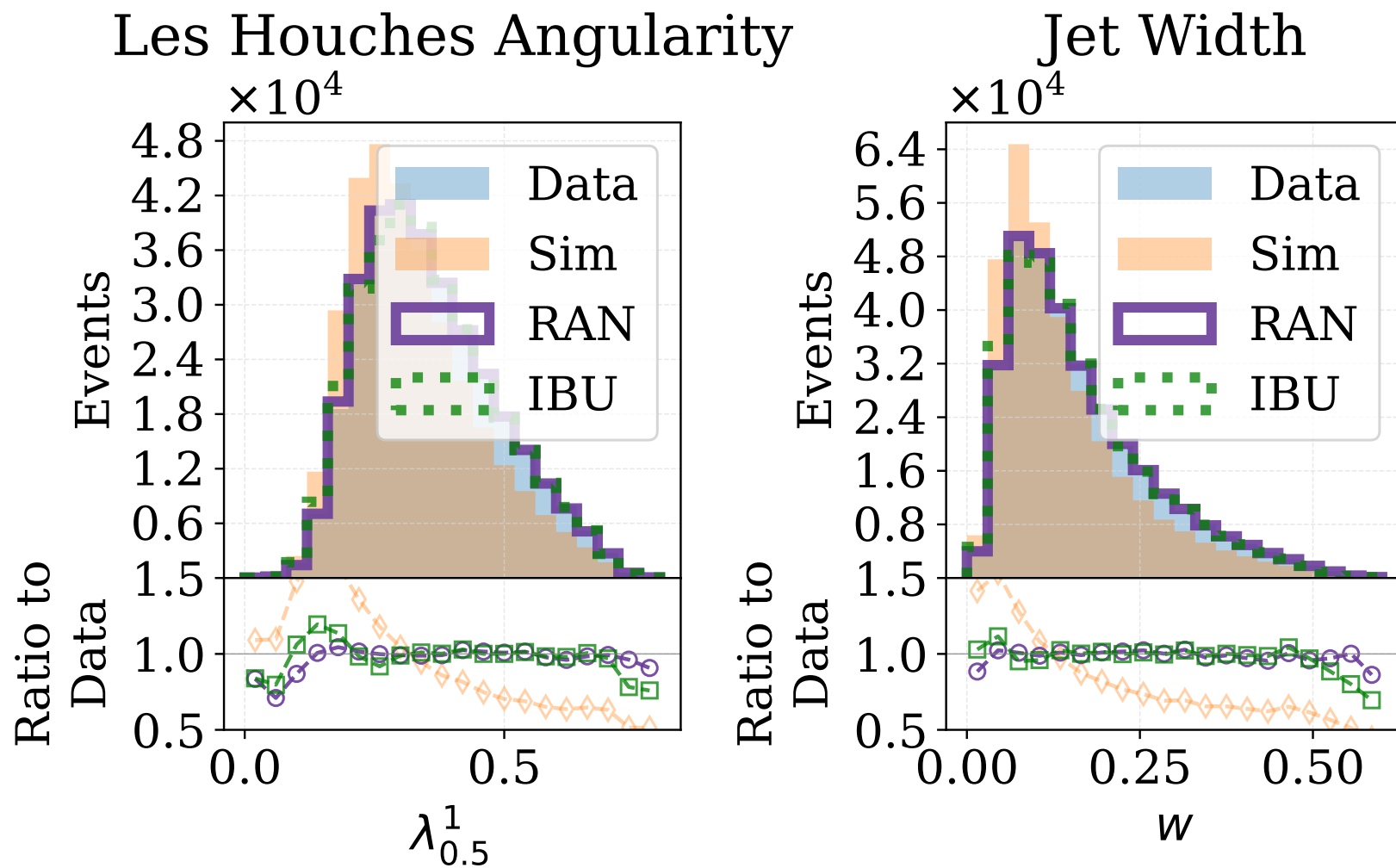
Observable	Gen	IBU	IBU impr. (%)	RAN	RAN impr. (%)
Mass and hard scale (IRC-safe kinematics)					
m [GeV]	7.809	45.93	−488.2	7.489	4.1
$\ln \rho$	39.27	6.889	82.5	0.8055	97.9
Continuous angularities (IRC-safe jet shapes)					
$\lambda_{0.5}^1$	48.98	8.014	83.6	0.8509	98.3
w	45.23	5.837	87.1	0.9044	98.0
λ_2^1	35.08	3.315	90.5	0.6502	98.1
Splitting and 2-prong substructure					
$z_g^\dagger$	8.086	8.086	0.0	0.6643	91.8
$\tau_{21}^{(\beta=1)}$	41.75	26.17	37.3	0.7443	98.2
Hadronization, multiplicity and fragmentation (IRC-unsafe)					
M	31.10	49.85	−60.3	1.441	95.4
n_{ch}	25.58	38.38	−50.1	1.333	94.8
f_{ch}	4.095	3.766	8.0	0.5547	86.5
p_T^D	33.05	16.05	51.4	0.9706	97.1
q	7.982	3.921	50.9	0.3415	95.7

[†] IBU's purity binning produced a single bin, so IBU failed to unfold.

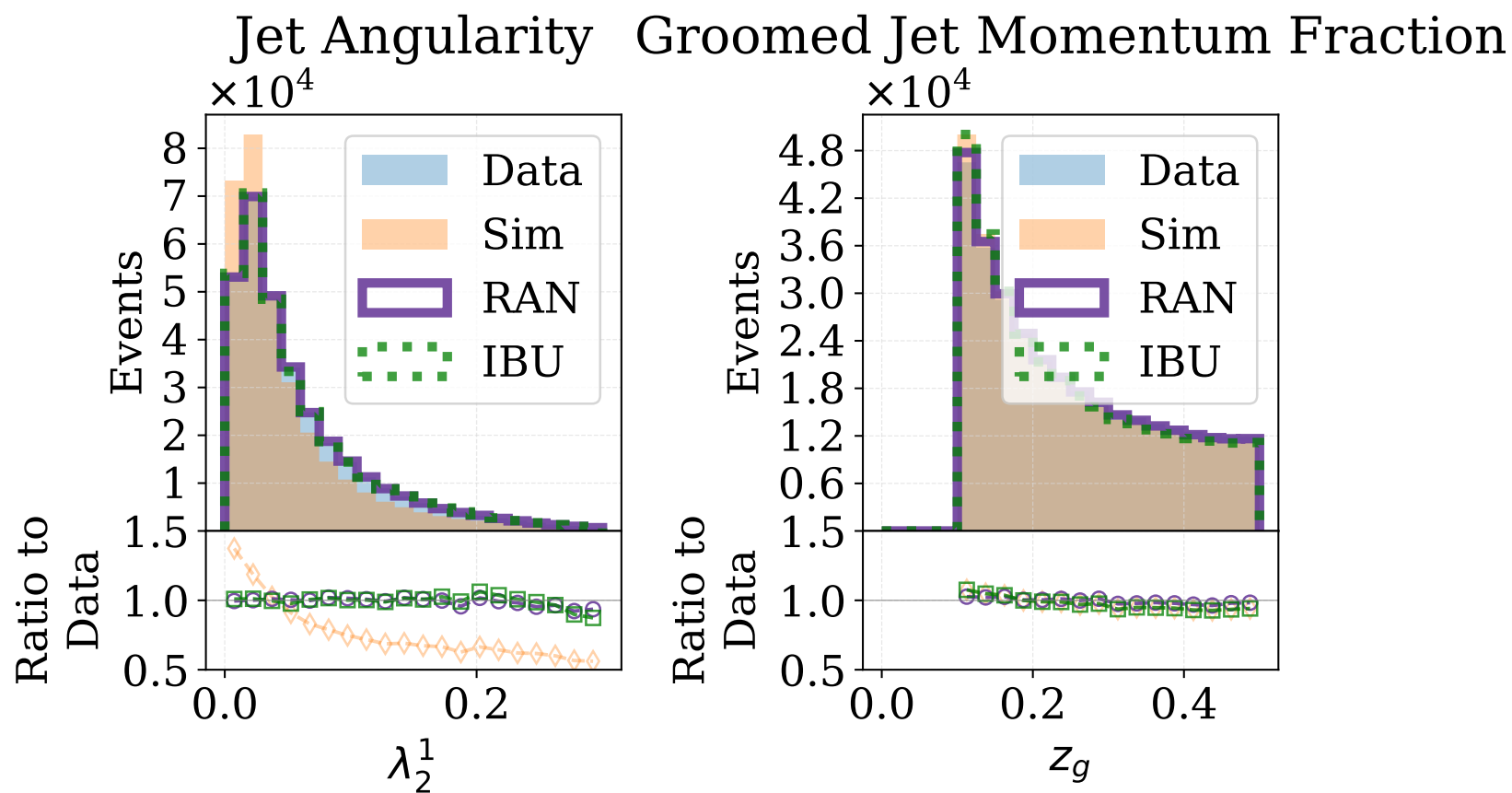
Detector Level (1 of 6)



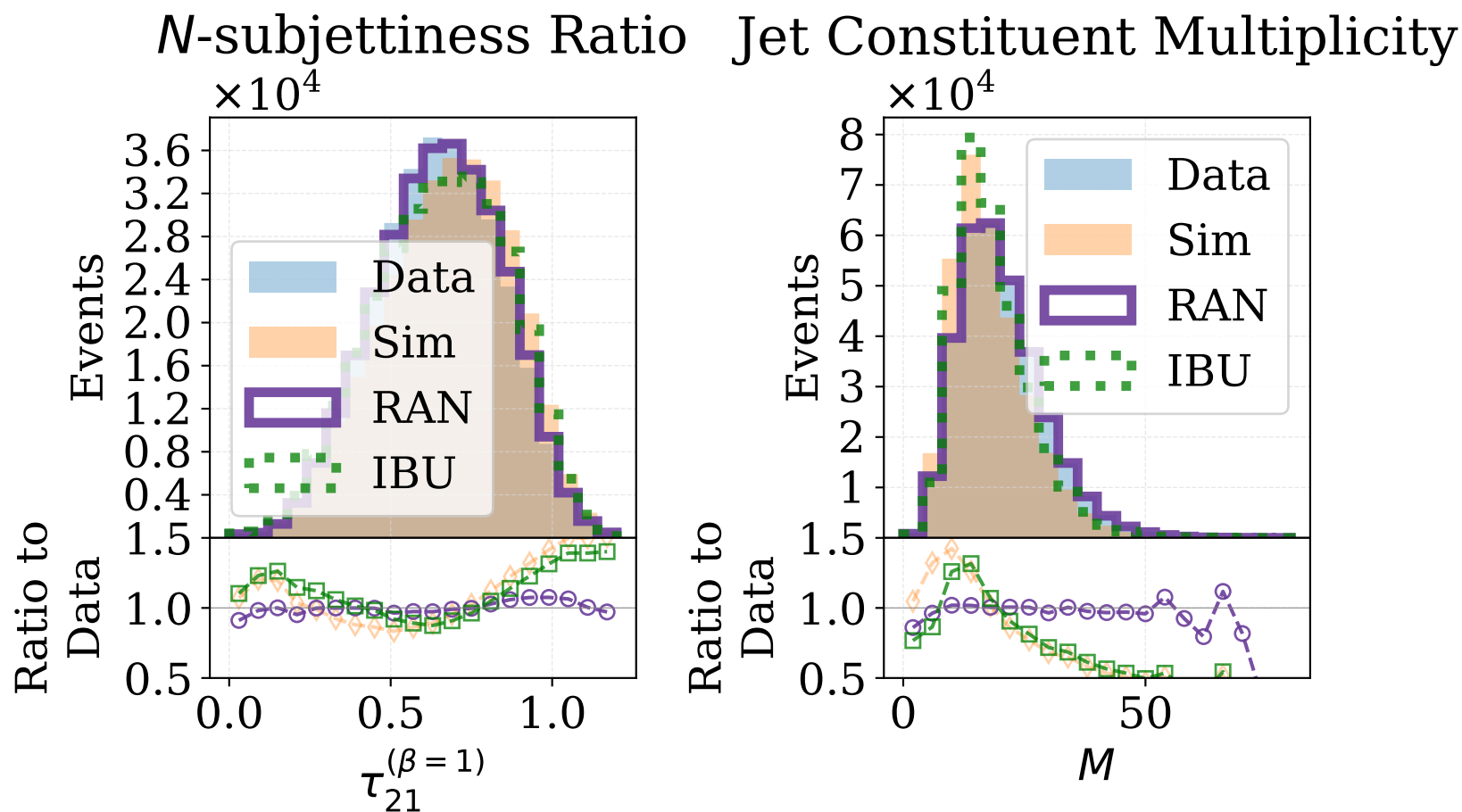
Detector Level (2 of 6)



Detector Level (3 of 6)

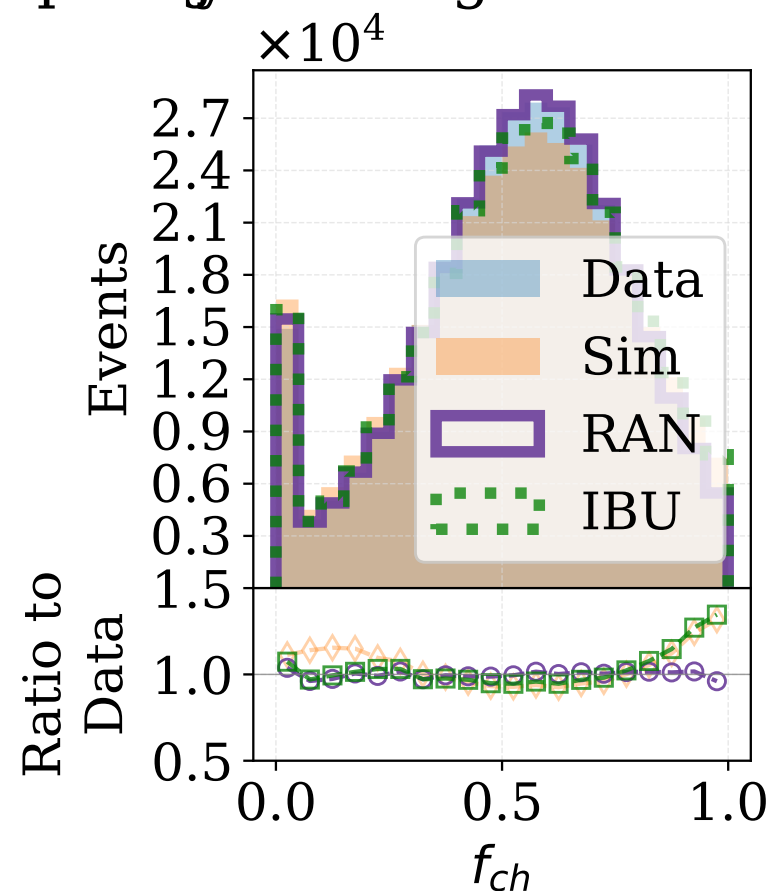
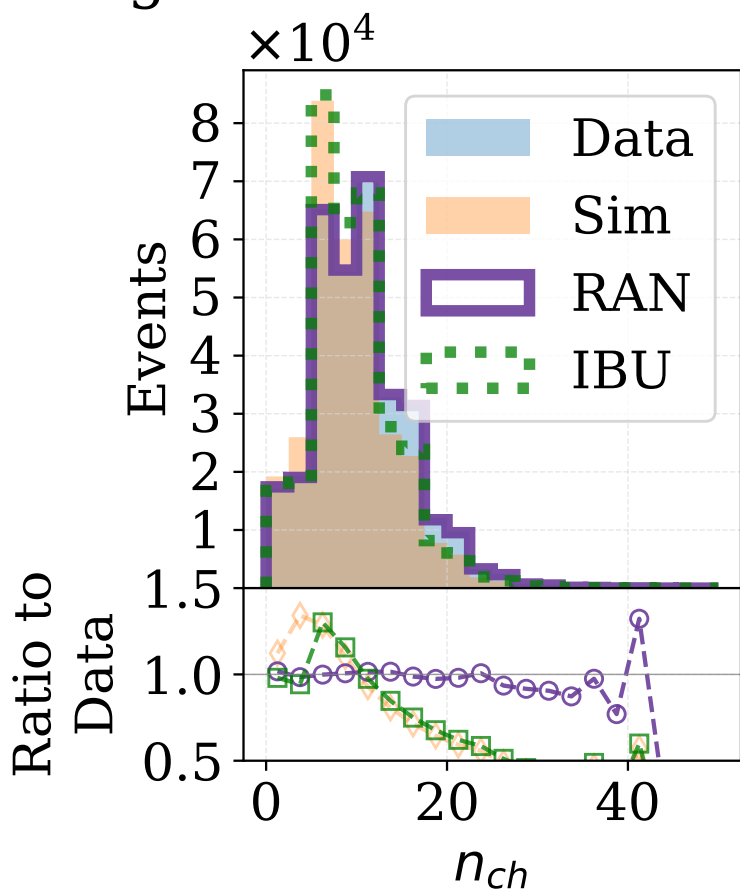


Detector Level (4 of 6)



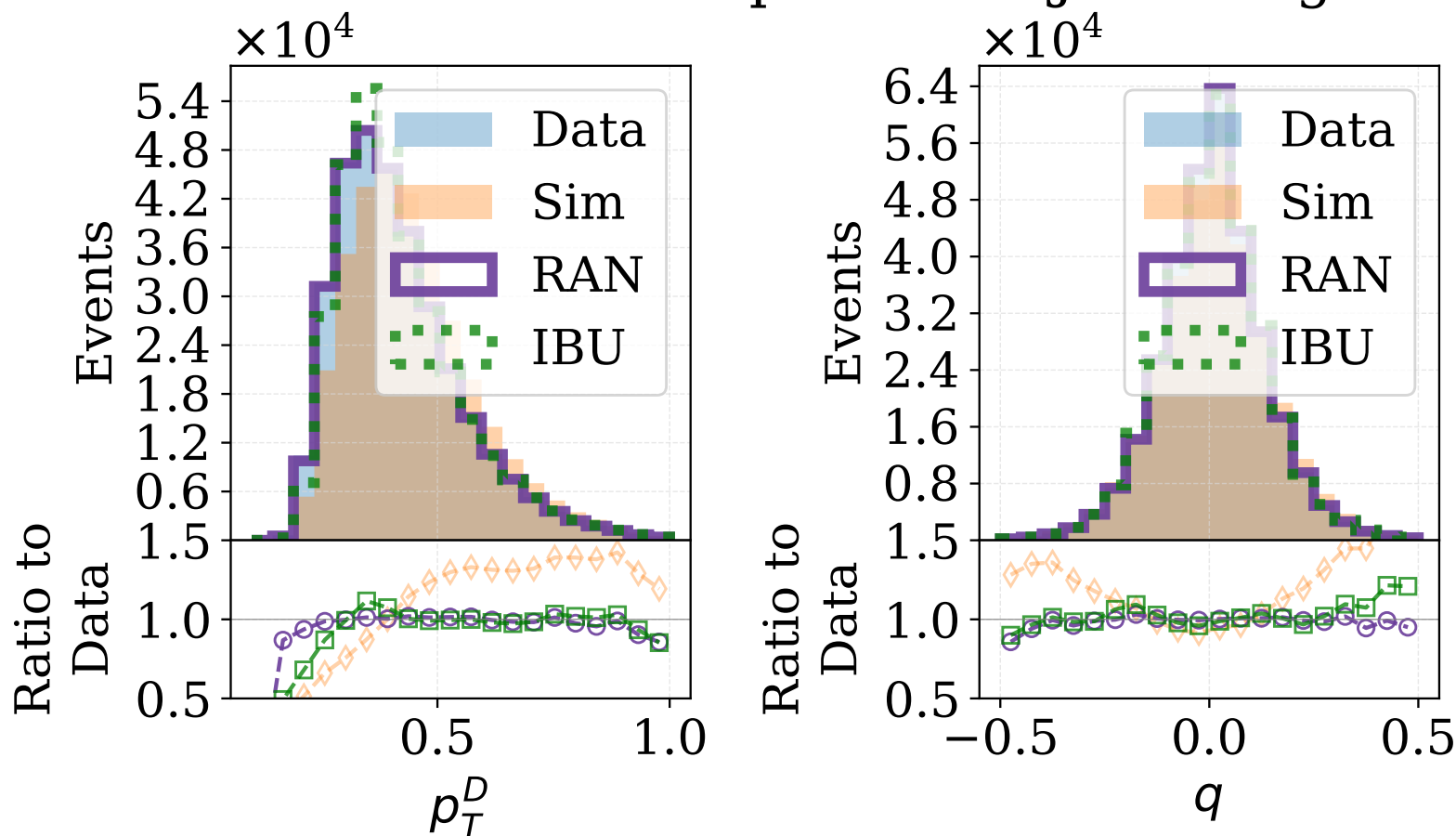
Detector Level (5 of 6)

Charged Constituent Multiplicity Jet Charge Fraction

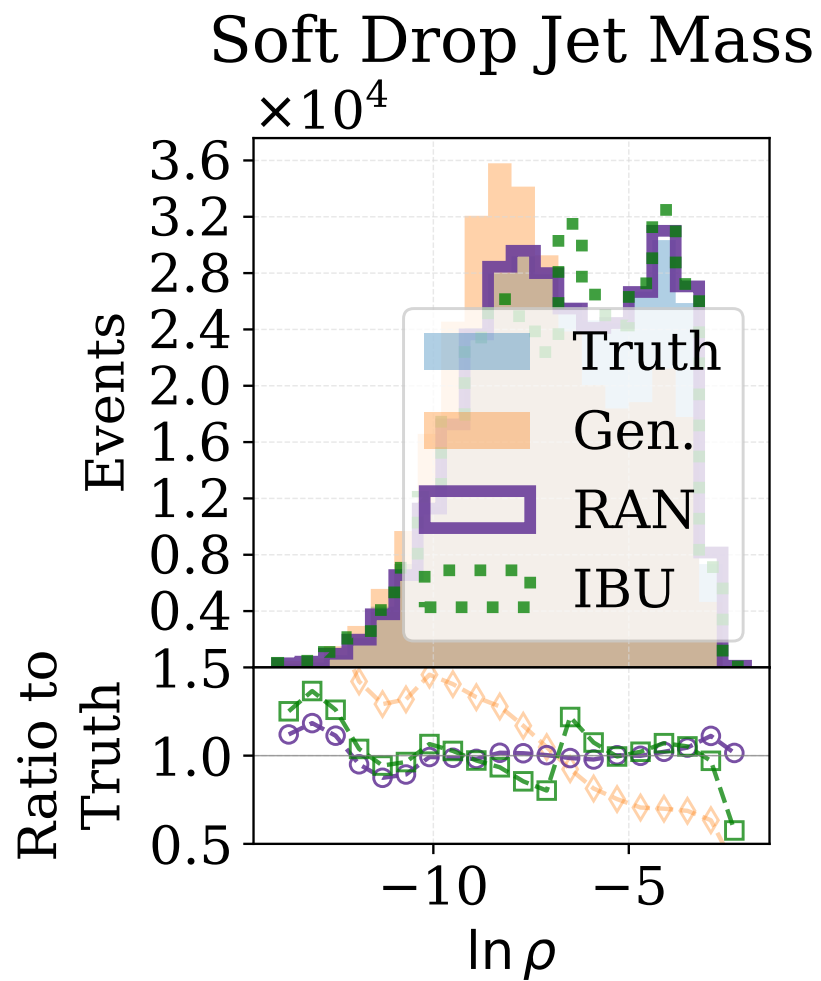
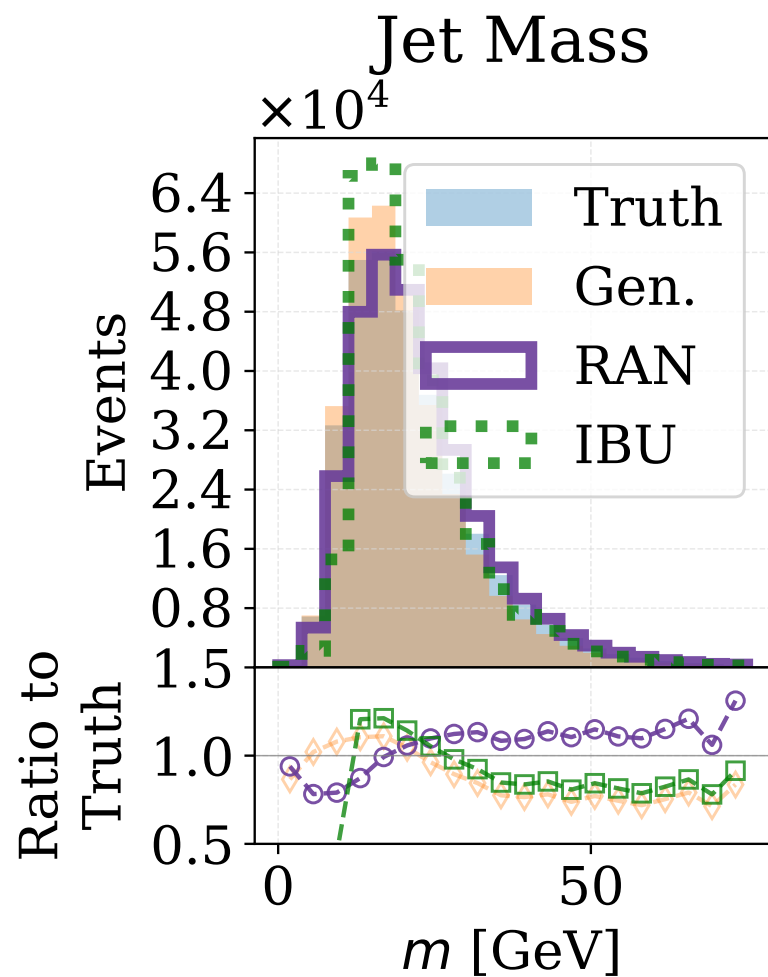


Detector Level (6 of 6)

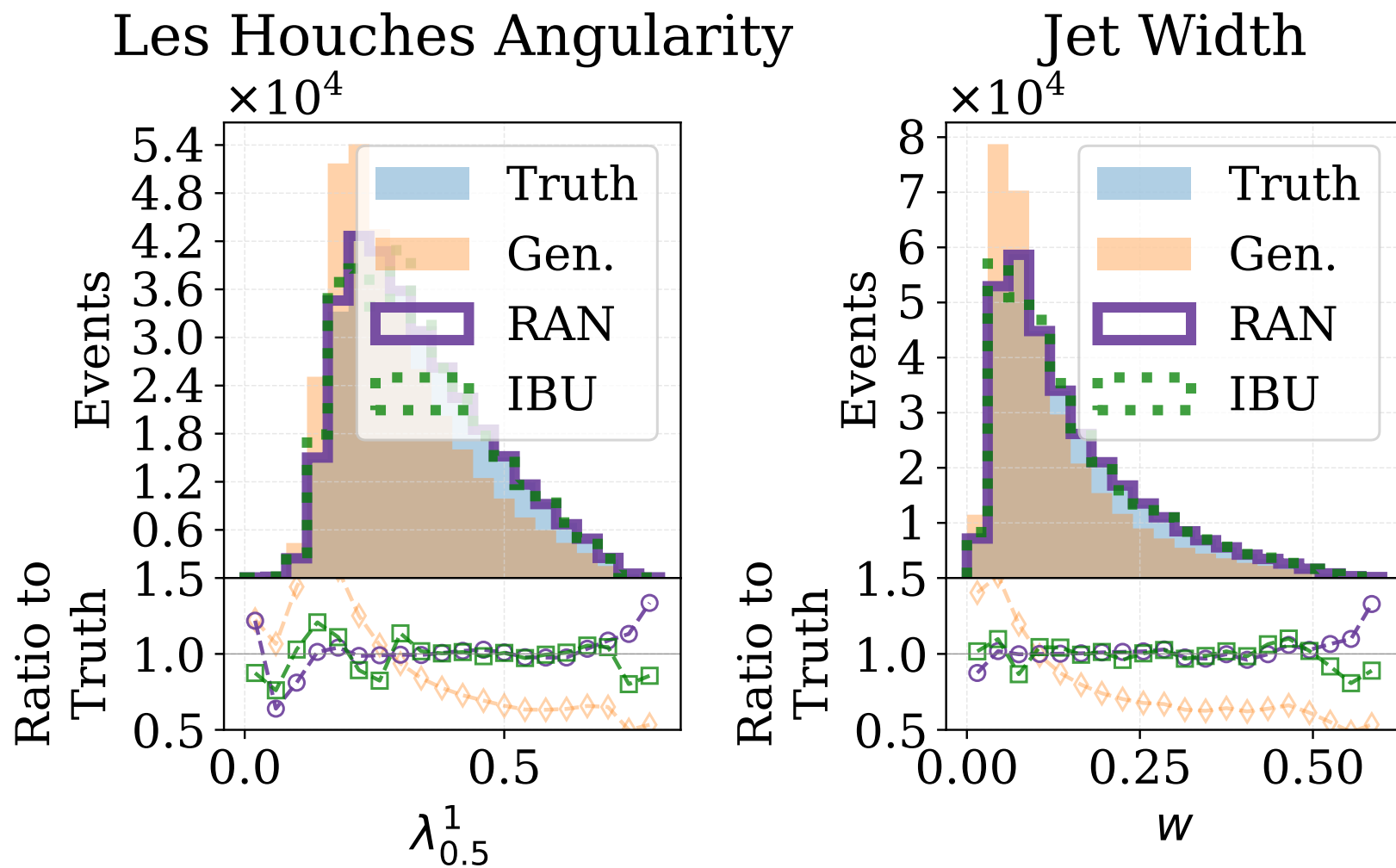
Transverse Momentum Dispersion Jet Charge



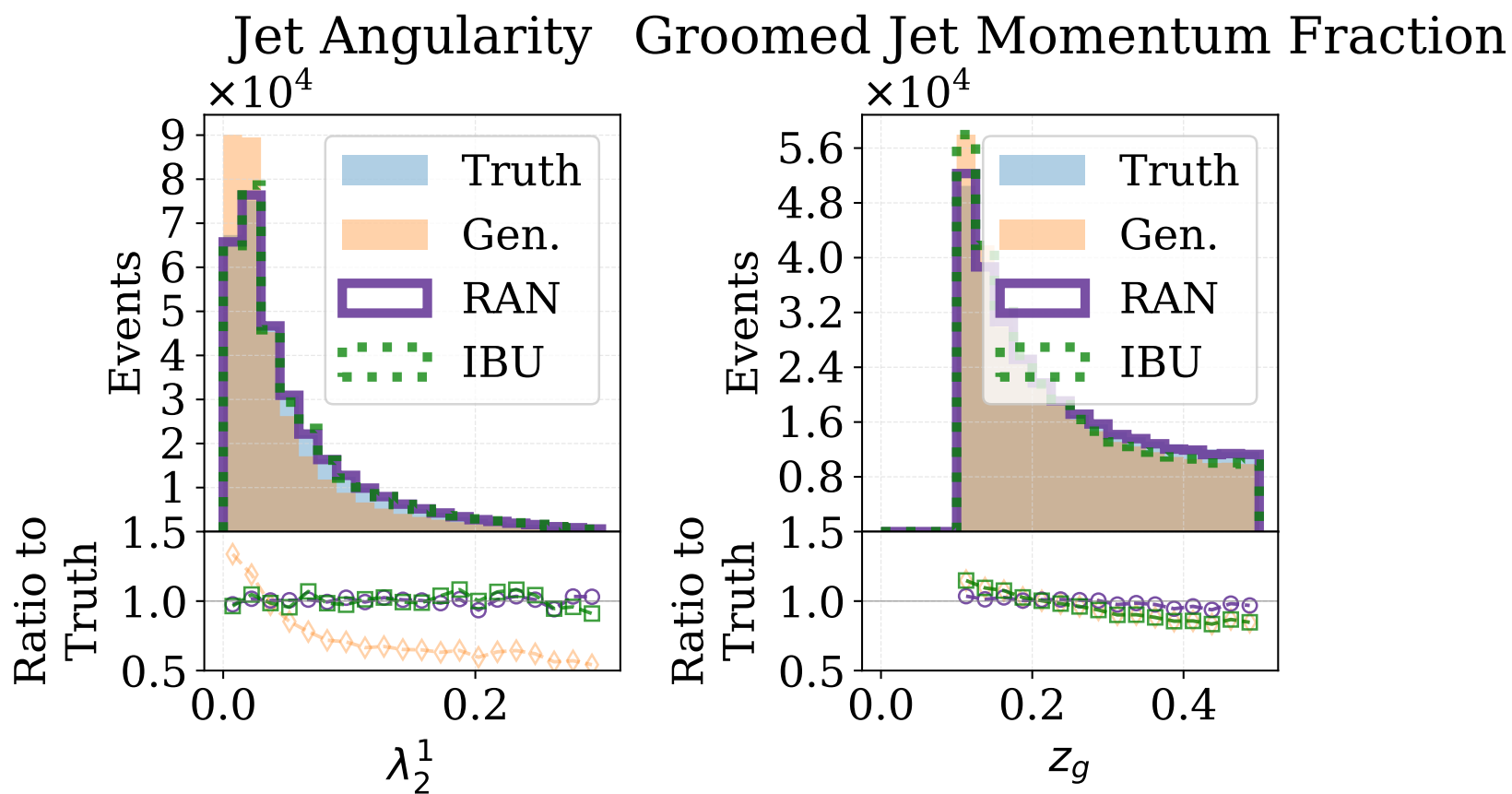
Particle Level (1 of 6)



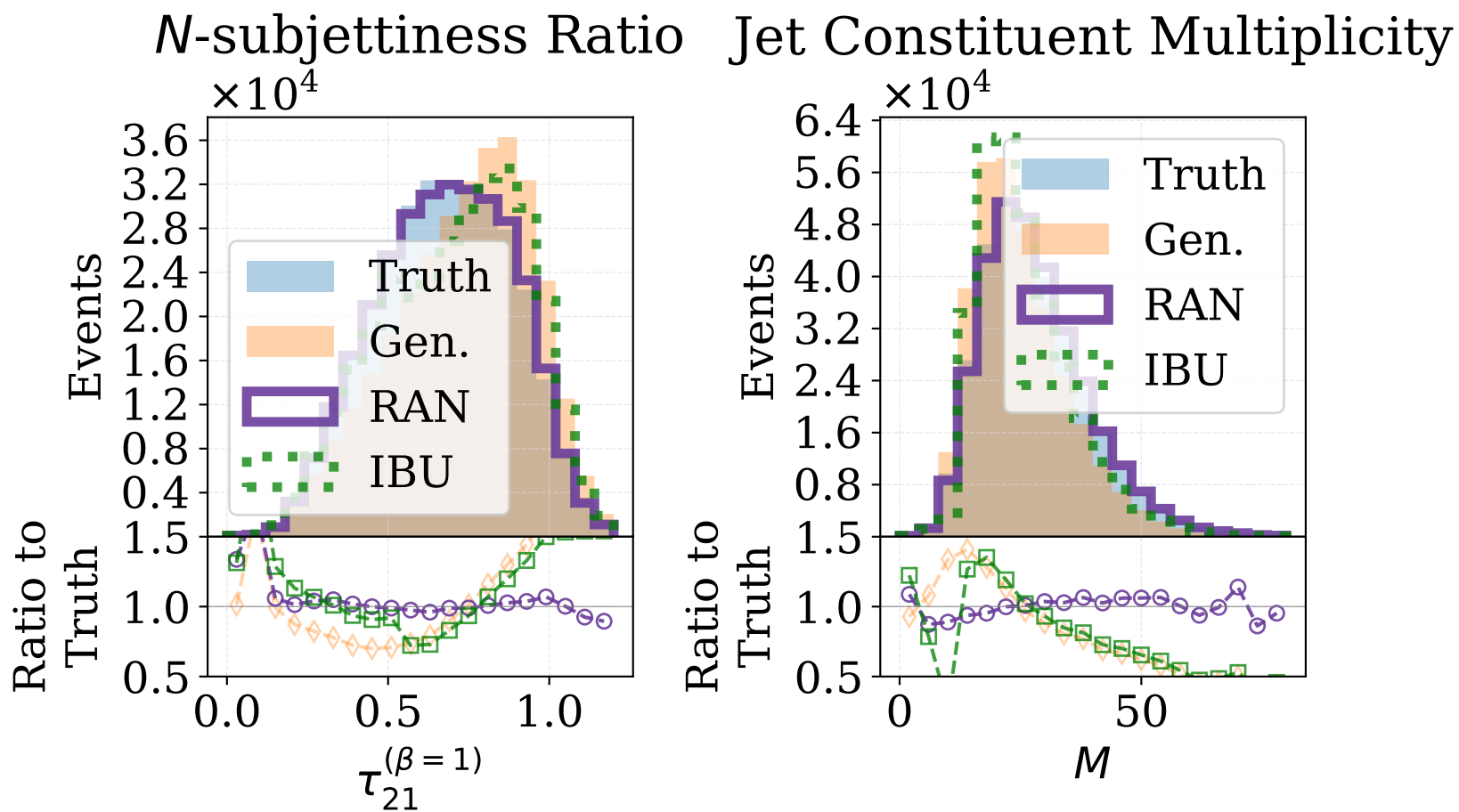
Particle Level (2 of 6)



Particle Level (3 of 6)

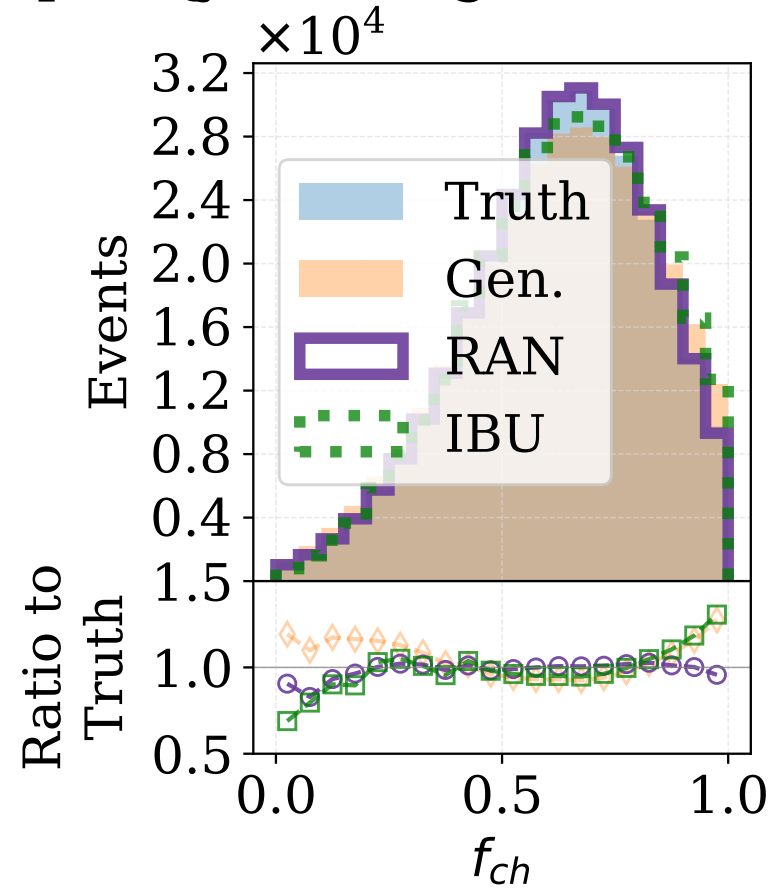
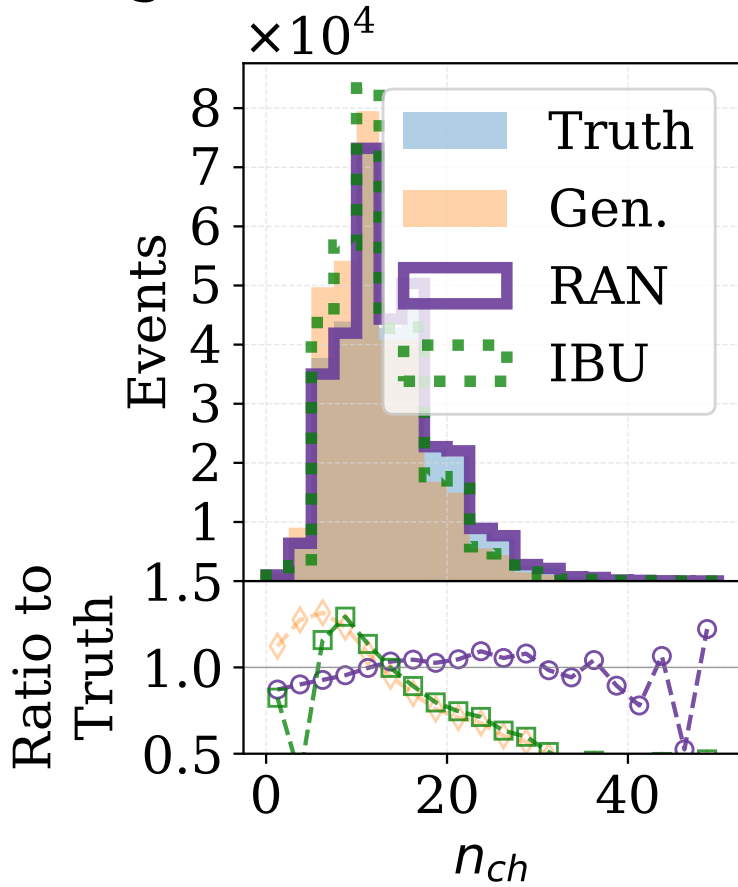


Particle Level (4 of 6)



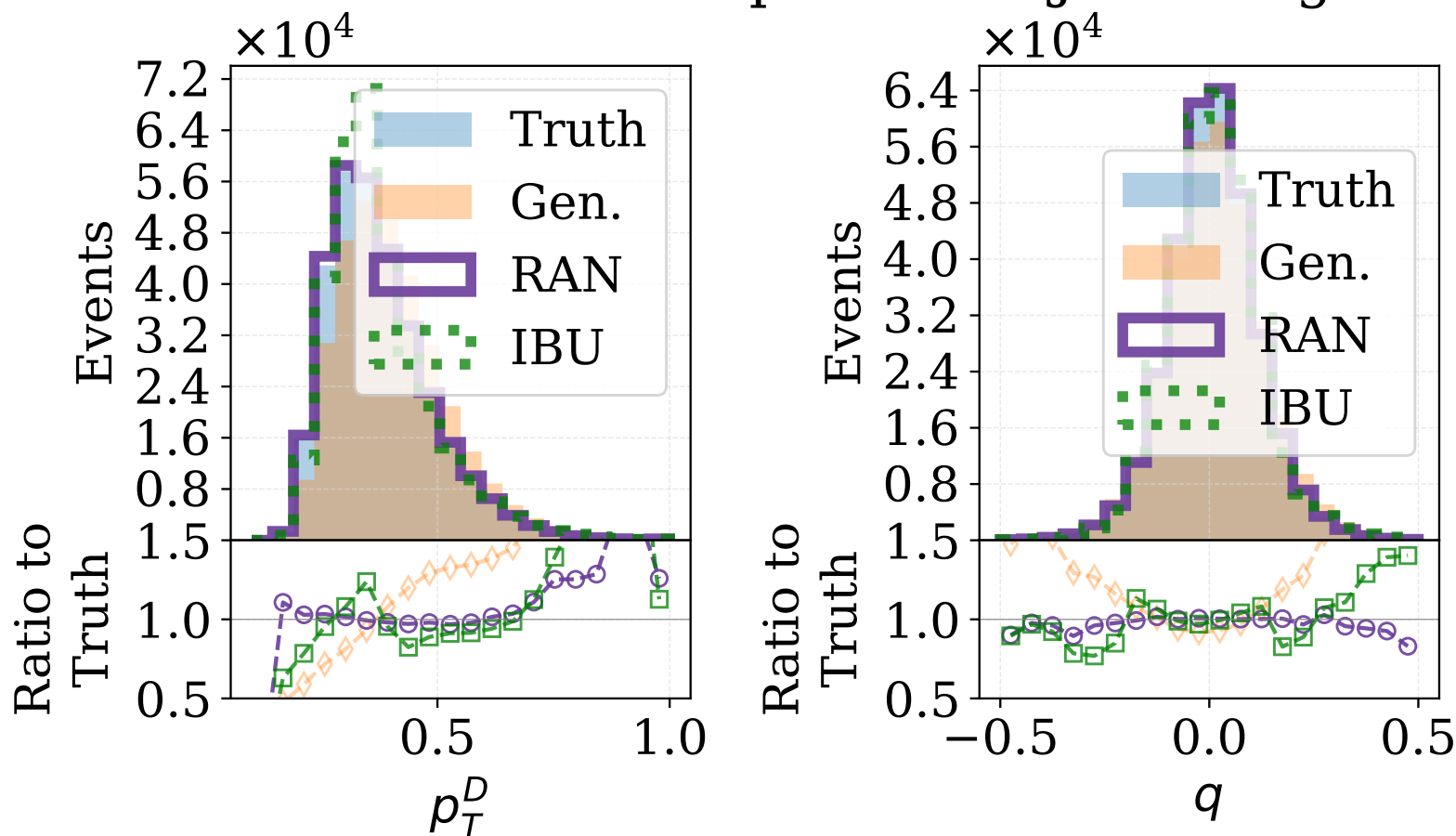
Particle Level (5 of 6)

Charged Constituent Multiplicity



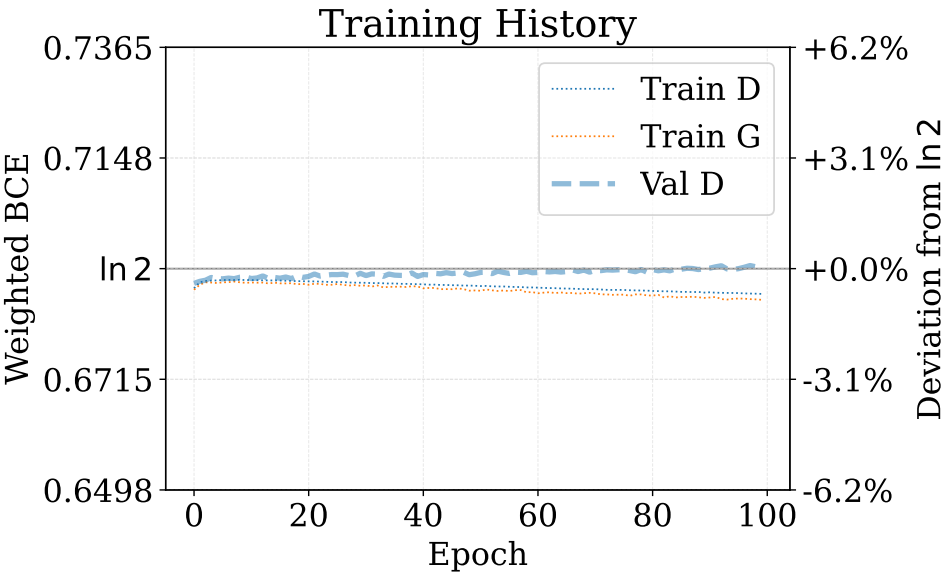
Particle Level (6 of 6)

Transverse Momentum Dispersion Jet Charge



Training diagnostics

Training losses



Selection criteria

